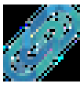
Exponential growth and decay 

1

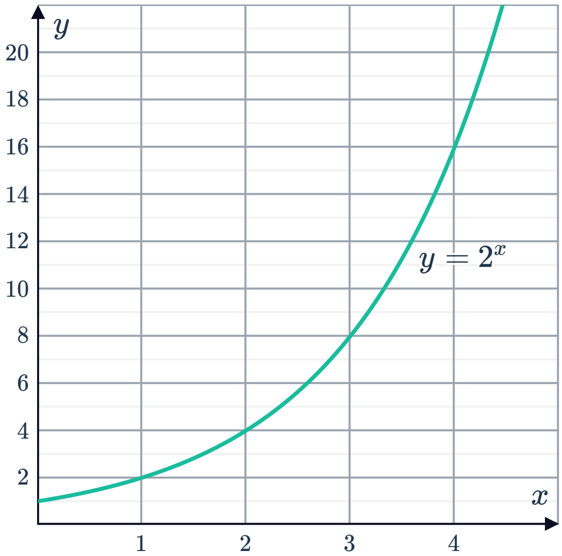
a

Number of hours passed (x)	0	1	2	3	4
Number of bacteria (y)	1	2	4	8	16

b $y = 2^x$

c $32\,768$

d



e The initial number of bacteria, in this case 1.

2

a

Day	0	1	2	3	4
Number of people infected	1	5	25	125	625

b 5^n

c 8 days

3

a

Month (x)	0	1	2	3
Amount remaining in kilograms (y)	1	$\frac{1}{5}$	$\frac{1}{25}$	$\frac{1}{125}$

b $y = 5^{-x}$

c $\frac{1}{78\,125}$ kg

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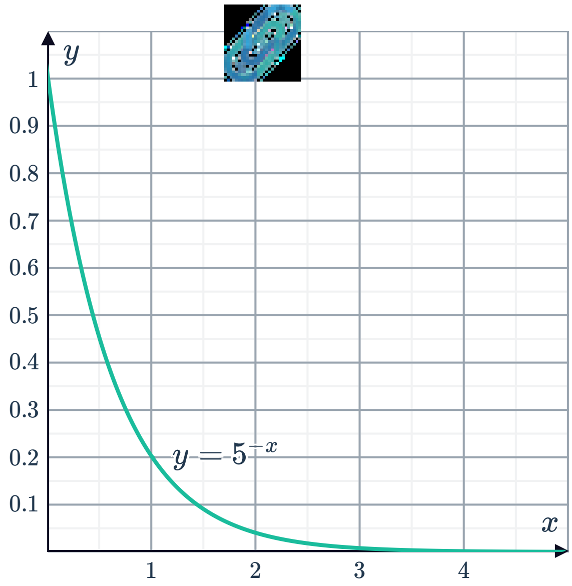
d



e The initial amount of isotope, in this case 1 kg.

4

a

Seconds passed (x)	0	1	2	3	4	5
Undissolved sugar in grams (y)	1	$\frac{1}{4}$	$\frac{1}{16}$	$\frac{1}{64}$	$\frac{1}{256}$	$\frac{1}{1024}$

b

i $y = \left(\frac{1}{4}\right)^t$

ii $y = 4^{-t}$

c $\frac{3}{16}$ g

d $\frac{3}{64}$ g

e It is decreasing at a decreasing rate.

f No, the model shows the amount of undissolved sugar is decreasing at a decreasing rate and has an asymptote of $y = 0$.

5

a 64

b $12\,288$

c $49\,152$

d The number of reactions is increasing at an increasing rate.

6

a $A = \left(\frac{1}{2}\right)^n$

b **A**

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7

a $y = (2)^x$

b That there is 1 layer of paper when no folds have been made.

c 1024

d 40.96 mm

8

a

Number of rounds (r)	3	4	5	6
Number of players (p)	8	16	32	64

b Any power of 2 such as $2, 4, 8, 16, 32, 64...$

c $p = 2^r$

d 1024

9

a 8

b 2^n

c 256

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